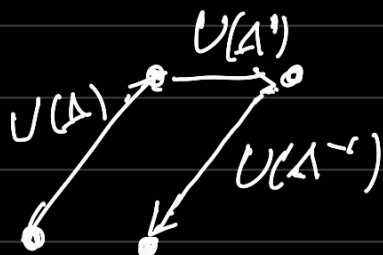


$$\nabla U(\Lambda' \Lambda) = U(\Lambda') U(\Lambda)$$

$$U(\mathbb{1} + \delta\omega) = \mathbb{1} + \frac{i}{2} \delta\omega_{\mu\nu} M^{\mu\nu}$$

- Look at the path



$$U(\Lambda^{-1}) U(\Lambda') U(\Lambda) = U(\Lambda^{-1} \Lambda' \Lambda)$$

$$\Lambda' \approx \mathbb{1} + \delta\omega'$$

$$\begin{aligned} \text{LHS)} \quad & U(\Lambda)^{-1} \left(\mathbb{1} + \frac{i}{2} \delta\omega'_{\mu\nu} M^{\mu\nu} \right) U(\Lambda) \\ &= U(\Lambda)^{-1} U(\Lambda) + \frac{i}{2} U(\Lambda)^{-1} \frac{i}{2} \delta\omega'_{\mu\nu} M^{\mu\nu} U(\Lambda) \\ &= \mathbb{1} + \frac{i}{2} U(\Lambda)^{-1} \frac{i}{2} \delta\omega'_{\mu\nu} M^{\mu\nu} U(\Lambda) \end{aligned}$$

$$\begin{aligned} \text{RHS)} \quad & U(\mathbb{1} + \Lambda^{-1} \delta\omega \Lambda) \\ &= \mathbb{1} + \frac{i}{2} (\Lambda^{-1} \delta\omega \Lambda)_{\rho\lambda} M^{\rho\lambda} \end{aligned}$$

so

$$U(\Lambda)^{-1} \delta\omega_{\mu\nu} M^{\mu\nu} U(\Lambda) = \Lambda_{\rho}^{\mu} \delta\omega_{\mu\nu} \Lambda_{\lambda}^{\nu} M^{\rho\lambda}$$

$\delta\omega_{\mu\nu}$ are constants

$$U(\Lambda)^{-1} \tilde{M}^{\mu\nu} U(\Lambda) = \Lambda_{\rho}^{\mu} \Lambda_{\sigma}^{\nu} M^{\rho\sigma}$$

Also, $U(\Lambda)^{-1} \tilde{M}^{\mu\nu} U(\Lambda)$, since

$$(\Lambda, a) \in \text{ISO}^+(1,3) \ltimes \mathbb{R}^{1,3}$$

$$U(\Lambda, a) \in \text{End}(\mathcal{H})$$

$$U(\Lambda) \equiv U(\Lambda, 0)$$

$$M^{\mu\nu} \equiv U(M, 0)^{\mu\nu}$$

$$\text{End}(\mathcal{H}) \cong \text{dx} \otimes \frac{\partial}{\partial x} \sim (1,1)$$

if the space on $\mathcal{H}$ has a metric.

(Auro Maresca)

(B. West)

$$J_i = \frac{1}{2} \epsilon_{ijk} M^{jk} \quad | \quad K_i = M^{0i}$$

$$\begin{cases} [J_i, J_j] = i\hbar \epsilon_{ijk} J_k \\ [J_i, K_j] = i\hbar \epsilon_{ijk} K_k & \text{(ii)} \\ [K_i, K_j] = -i\hbar \epsilon_{ijk} J_k & \text{(iii)} \end{cases}$$

(ii) Implies $\vec{K}$ transforms as a 3-vector

(iii) Series of boosts is eq. to a Rotation.

$$\langle x, t | \hat{\phi}(x, 0) | x, t \rangle$$

$$\begin{aligned} i\partial_t |\psi\rangle &= \hat{H} |\psi\rangle \\ \text{"formally"} \quad |\psi(t)\rangle &= e^{-i\hat{H}t} |\psi_0\rangle \\ \rightarrow \langle x | e^{i\hat{H}t} \hat{\phi}(x, 0) e^{-i\hat{H}t} | x \rangle \end{aligned}$$

• Hersenberg:

$$\hat{\phi}(x, t) = e^{i\hat{H}t} \hat{\phi}(x, 0) e^{-i\hat{H}t}$$

$$\hookrightarrow \phi(\vec{x}, 0) = e^{-i p_\mu x_\mu} \phi(0, 0) e^{+i p_\mu x_\mu}$$

• Already learned, $\mathbb{R}^{1,3}$:

$$U(1, a) = e^{-i p_\mu a_\mu}$$

$$\begin{aligned} U(1, a) |p\rangle &= e^{-i p_\mu a_\mu} |p\rangle \\ \langle x | U(1, a) |p\rangle &= e^{-i p_\mu a_\mu} \langle x | p \rangle \\ &= e^{-i p_\mu a_\mu} e^{-i p_\mu x_\mu} \\ &= \exp(-i p_\mu (x + a)) \\ \downarrow \\ \langle 0 | U(1, a) |p\rangle &= e^{-i p_\mu a_\mu} \end{aligned}$$

$$\rightarrow U(1, a) \int dx |x\rangle \langle x| p\rangle = \int dx |x\rangle \bar{e} \langle x| p\rangle$$

$$\int dx U(1, a) |x\rangle \bar{e} = \int dx \bar{e}^{ixp} e^{ip^\mu a_\mu} |x\rangle$$

$$U(1, a) |x\rangle = e^{ip^\mu a_\mu} |x\rangle$$

$$|x\rangle = e^{+ip^\mu x_\mu} |0\rangle$$

Thus,

$$e^{-ipx} \phi(0) e^{ipx} = \phi(x)$$

Define: Spacetime translation operator:

$$T(a) \equiv U(1, a) \equiv \exp[-i \tilde{P}^\mu a_\mu]$$

$$T^{-1}(a) \phi(x) T(a)$$

$$= e^{ip^\mu a_\mu} [e^{-ipx} \phi(0) e^{ipx}] e^{-ipa}$$

$$= \exp(-i P^\mu (x_\mu - a_\mu)) \phi(0) \exp(i P^\mu (x_\mu - a_\mu))$$

Thus

$$T^{-1}(a) \phi(x) T(a) = \phi(x - a)$$

* Spacetime translations 'sandwich' $T_a^{-1} \phi T_a$

$$T(\delta a) = 1 - i \delta a_\mu P^\mu$$

$$U(\Lambda)^{-1} \phi(x) U(\Lambda) = \phi(\Lambda^{-1} x)$$

$$U(\Lambda)^{-1} \partial^\mu \phi(x) U(\Lambda) = \frac{\partial \bar{x}^\rho}{\partial x^\mu} \bar{\partial}^\rho \phi(\Lambda^{-1} x)$$

$$\bar{x}^\rho \equiv \Lambda^{-1 \rho}_\sigma x^\sigma$$

$$\partial^\mu \bar{x}^\rho = \partial^\mu \Lambda^\rho_\sigma x^\sigma \Rightarrow$$

$$U(\Lambda)^{-1} \partial^\mu \varphi(x) U(\Lambda) = (\Lambda^{-1})^\mu_\rho \bar{\partial}^\rho \varphi(\Lambda^{-1}x)$$

$$U(\Lambda)^{-1} \partial^2 \varphi(x) U(\Lambda) = \bar{\partial}^2 \varphi(\Lambda^{-1}x)$$

$$* \quad U(\Lambda)^{-1} \partial^\mu \varphi(x) U(\Lambda) = (\Lambda^{-1})^\mu_\rho \bar{\partial}^\rho \varphi(\Lambda^{-1}x)$$

let

$$\Lambda^{-1} \equiv \tilde{\Lambda}$$

$$U(\tilde{\Lambda}) \partial^\mu \varphi(x) U(\tilde{\Lambda})^{-1} = \tilde{\Lambda}^\mu_\rho \bar{\partial}^\rho \varphi(\tilde{\Lambda}x)$$